

Homework-Games Set 3. Total 5 problems

Homework-Games1: branching factor in chess play.

Suppose it is now "red" player's turn. See board situation on right figure. Our task is to count the number of moves he (red player) has to search and evaluate (he chooses the move with the best score!).

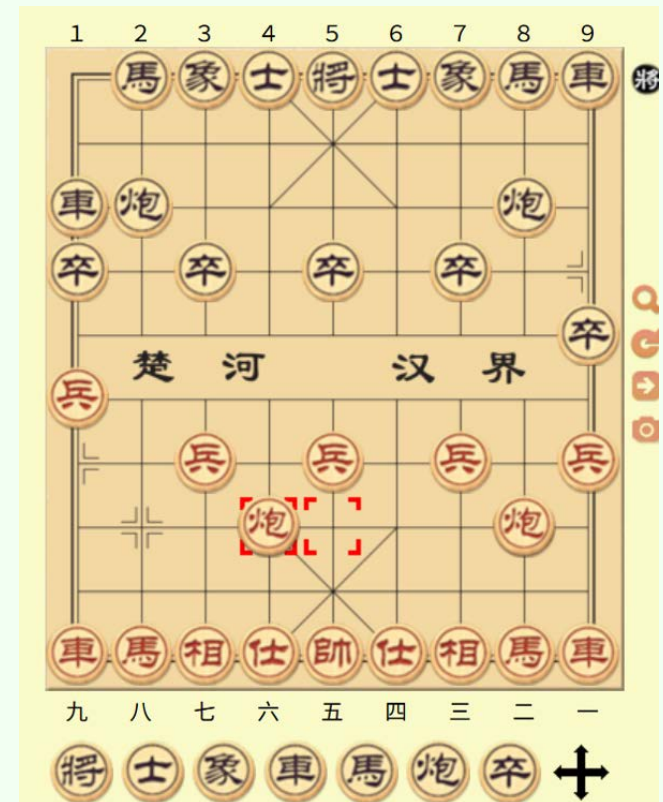
(A) How many different/distinct moves he can make?
Write down details for every pieces (棋子). For instance,
5个兵(soldier), each has 1 move
Left 炮(cannon) has $3+3+1+6=13$ moves

...

Add up number of moves for each of the 16 棋子。
This value is the number of distinct moves he can do, or the number of adjacent nodes on the chess board situation graph. In tree search type algorithm, this is the number of choices, or branching factor.

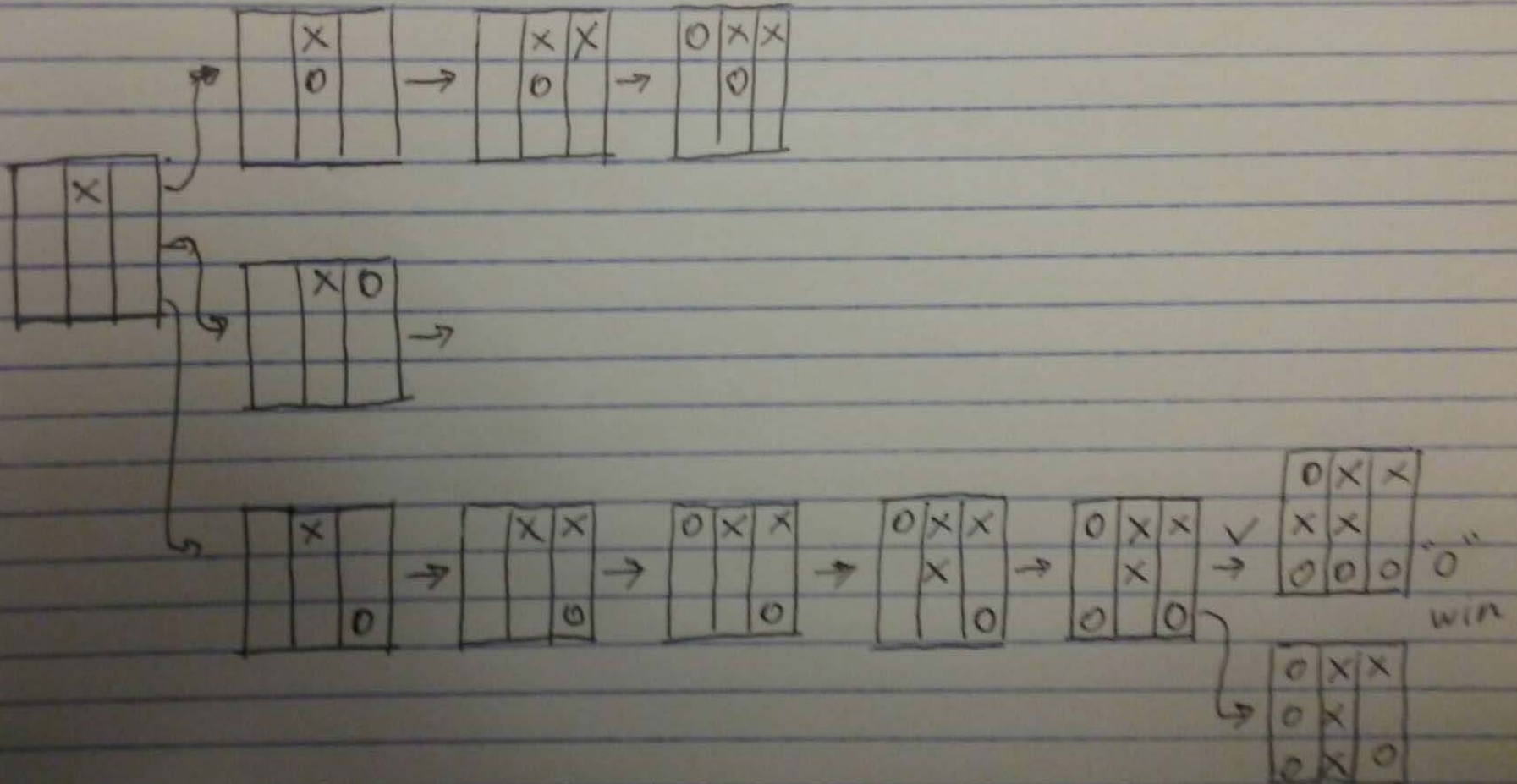
(B) If he use 1-step look-ahead strategy (next step), how many evaluation of heuristic (assessment of the board situation) score he needs to do?

(C) If he use 3-step look-ahead strategy, how many evaluation of heuristic scores he needs to do? (assume the branching factor remains the same).

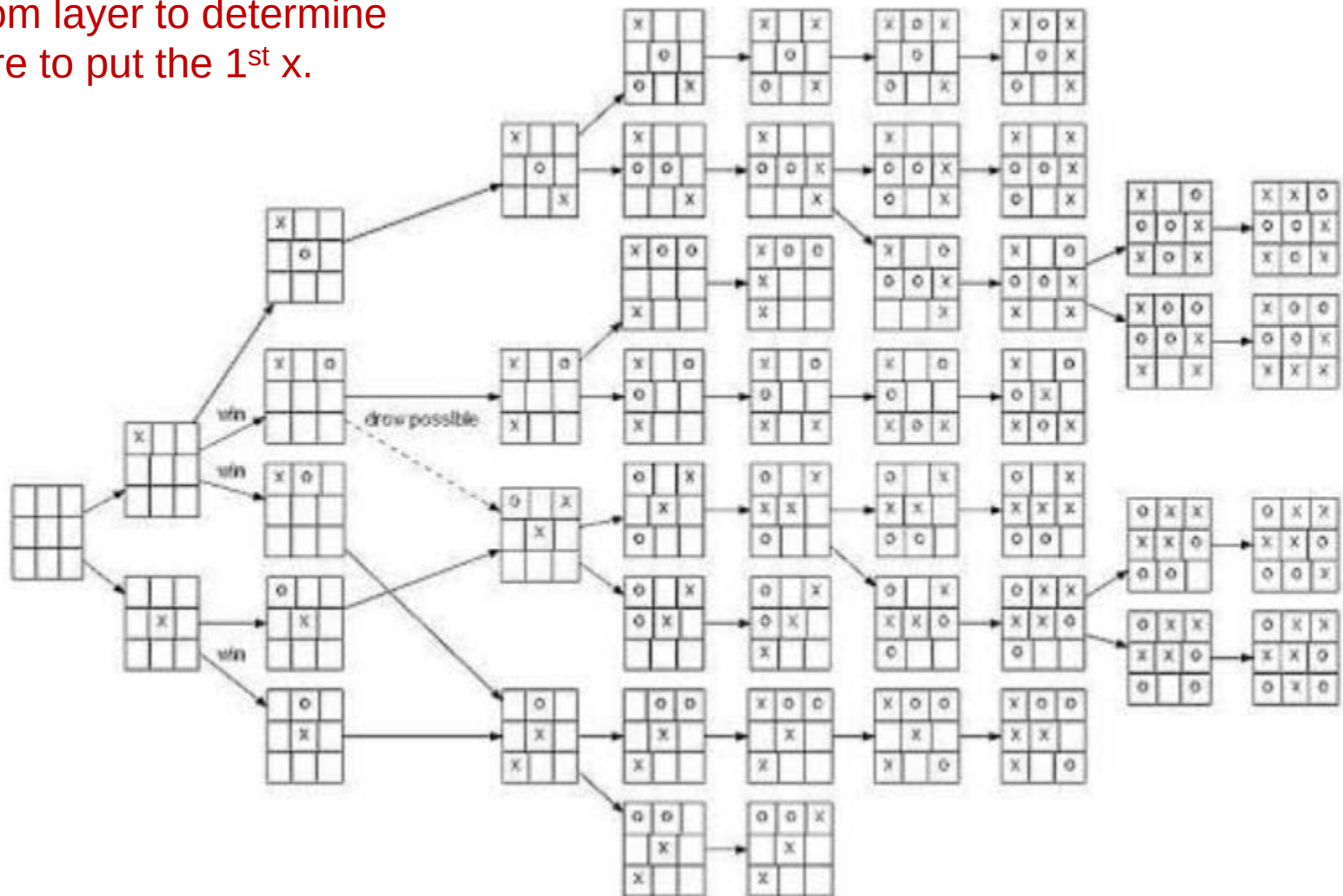


Homework-Game2: Complete the rest steps of the top two incomplete game plays using your own instincts.

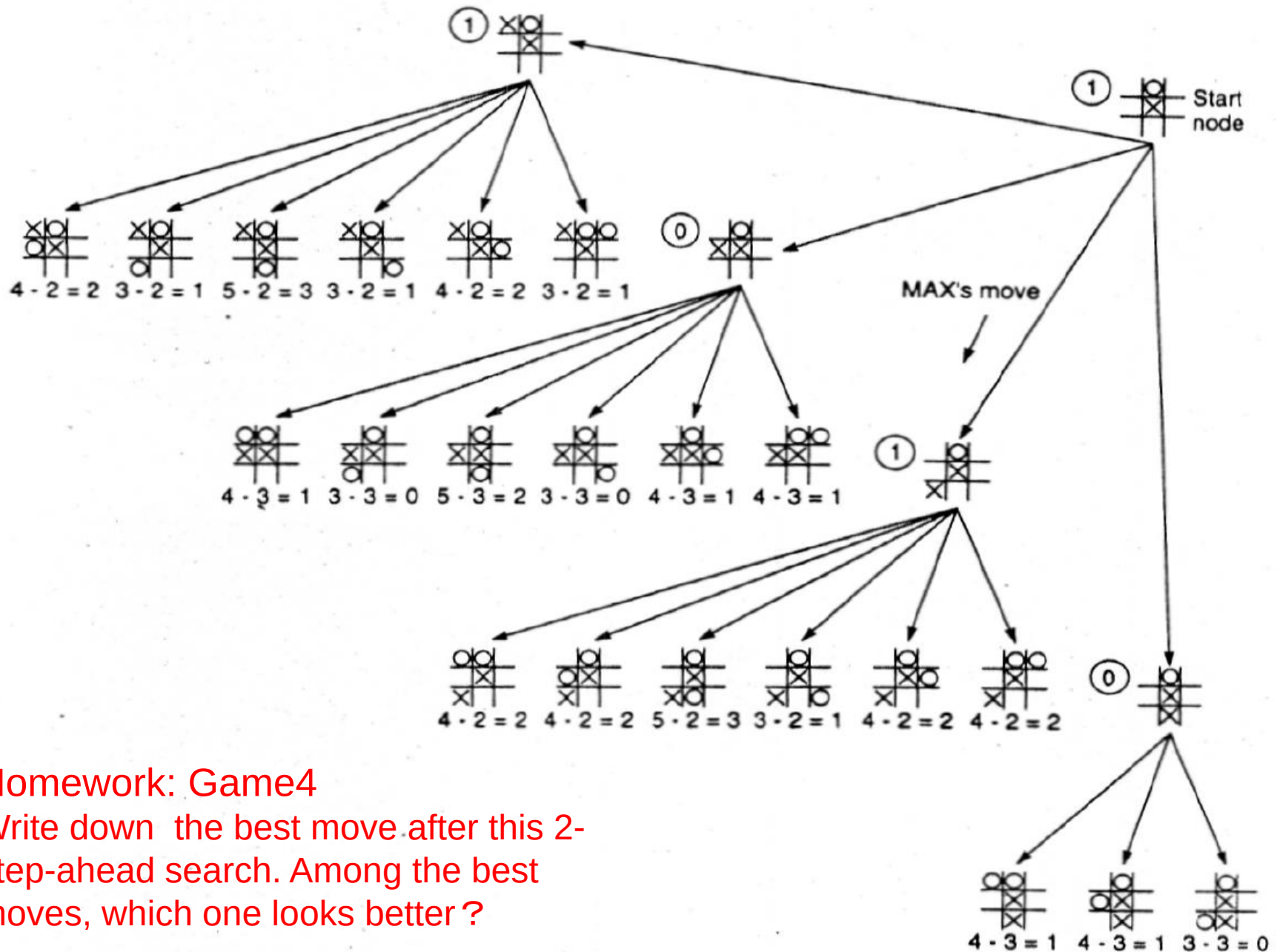
Homework: You complete the rest/remaining games



Do value backup from the bottom layer to determine where to put the 1st x.

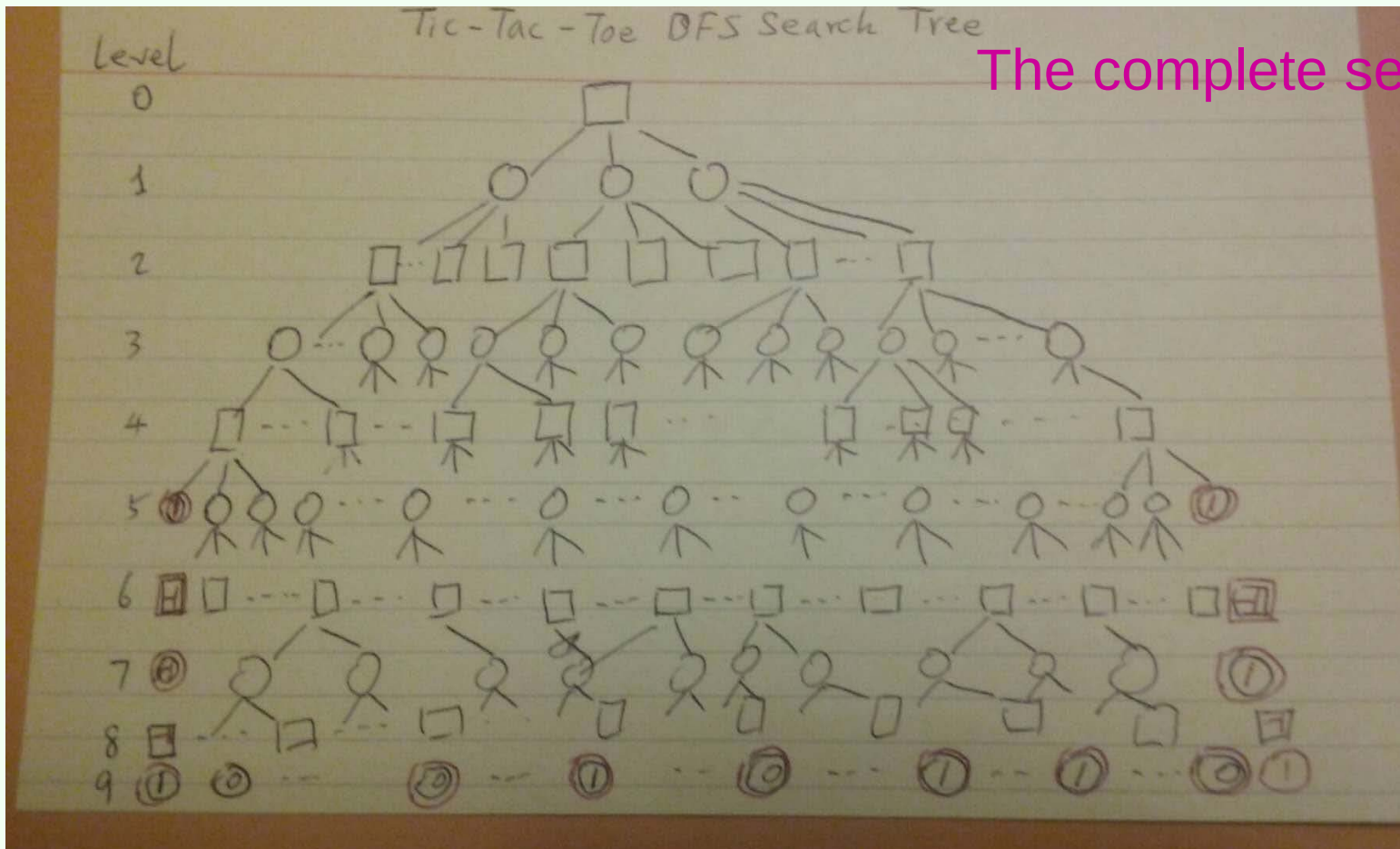


2-step-ahead search using heuristic evaluation function



Homework: Game4

Write down the best move after this 2-step-ahead search. Among the best moves, which one looks better?



The complete search tree

Tree property:

1. Leaf nodes occur at levels 5,6,7,8,9. Why?
2. Leaf nodes on level 5 has value +1. Why?
3. Leaf nodes on level 6 have value -1. Why?
4. Leaf nodes on level 9 (bottom) have values 0 or 1. Why?
5. How many leaf nodes on level 5?
6. How many leaf nodes on level 6? (optional)
7. How many leaf nodes on level 9? (optional)

HOMEWORK-
Game5:
Explain these
properties
(See next page)

Homework: Game5. This tree is the same search tree in previous page

The complete search tree

